



Hope you cover all key concept list if any missing

Now CCDE design scenarios (e.g., "Design Provider network with SR-EPE, BGP-LS, and PCE").

Real word case

Also Migration to SR from MPLS enable nw comple scinario

Here are comprehensive CCDE-level design scenarios with real-world cases, including a complete migration from MPLS/LDP to Segment Routing. These are designed for senior network architects with 16+ years of experience.

☒ CCDE Design Scenario 1: Design Provider Network with SR-EPE, BGP-LS, and PCE

☒ Scenario Overview

Company: Global Service Provider (GSP)

Network Size: 500 routers, 100 edge routers, 20 ISP peers

Requirements:

- Optimize egress traffic to specific ISP peers (cost optimization)
- Centralized TE for low-latency paths
- BGP-free core (core only runs IGP, not full BGP)
- Sub-50ms failover for all critical paths

☒ Design Architecture

Layer 1: Core Network (IGP Domain)

```
Core Routers (50 routers):  
├─ OSPF/IS-IS Area 0 (SR-enabled)  
├─ SRGB: [16000-23999]  
├─ Node SIDs: 101-150 (per core router)  
├─ Flex-Algo 128: Low-latency path  
├─ TI-LFA: <50ms failover
```

Key Design Points:

- SR-MPLS enabled on all core routers

- **Node SIDs** for each router (global uniqueness)
- **Flex- Algo 128** for low-latency traffic (video/VoIP)
- **TI-LFA** for fast reroute (no RSTP/RSVP-TE needed)

Layer 2: Edge Network (Edge Routers + BGP EPE)

Edge Routers (100 routers):

- └─ BGP MP (multiprotocol) with ISP peers
- └─ BGP EPE SIDs enabled (peer-level segments)
- └─ BGP-LS enabled (advertise topology to PCE)
- └─ SR Policy headend (steers traffic based on PCE)

BGP EPE Configuration (Cisco IOS XR):

```
router bgp 65000
  bgp router-id 10.0.0.1
  !
  bgp bestpath egress-peer-engineering enable
  !
  neighbor 203.0.113.1
    remote-as 65100 ! ISP-1
    description ISP-1-Peering
    !
    egress-peer-engineering
      peer-type external
      segment-routing
        peer-sid 32001 ! EPE SID for ISP-1
    !
  !
  !
  !
  neighbor 198.51.100.1
    remote-as 65200 ! ISP-2
    egress-peer-engineering
      peer-type external
      segment-routing
        peer-sid 32002 ! EPE SID for ISP-2
    !
  !
  !
```

How EPE Works:

Traffic to 8.8.8.8 (Google):

- Headend sees multiple exit paths (ISP-1, ISP-2)
- PCE computes: ISP-1 is cheaper (cost optimization)
- SR Policy: [Node-Edge-A, Peer-SID-ISP-1]
- Traffic goes via ISP-1 (not ISP-2)

Layer 3: SDN Controller (PCE + BGP-LS)

- SDN Controller (2 nodes for HA):
- └─ PCE (Path Computation Element)
 - └─ BGP-LS Collector (receives topology from all routers)
 - └─ ODN templates (automatic SR Policy creation)
 - └─ PCEP server (programs SR Policies into headends)
 - └─ RestAPI (northbound for applications)

BGP-LS Configuration (All Routers):

```
router bgp 65000
  bgp router-id 10.0.0.1
  !
  bgp link-state
    report all
  !
  !
  neighbor 10.0.100.1 ! SDN Controller
    remote-as 65000
    address-family link-state link-state
  !
  !
  !
```

What BGP-LS Advertises:

Information	Example
Nodes	Router IDs, loopbacks (10.0.0.1/32, 10.0.0.2/32)
Links	Interfaces (192.168.1.0/30), bandwidth (10G)
Prefixes	IP prefixes + Node SIDs (16101, 16102)
SR Capabilities	SRGB [16000-23999], Flex-Algo 128
EPE SIDs	Peer-SID 32001 (ISP-1), 32002 (ISP-2)

Layer 4: SR Policy Programming (PCEP)

PCEP Configuration (Edge Routers):

```
router segment-routing
  pce
    pce-name SDN-Controller
    address 10.0.100.1
    password PCEP-SECRET
  !
  stateful
  active
```

!
!

How PCE Programs SR Policy:

```
Edge Router A requests path to 8.8.8.8 (ODN template matches):
PCE computes: [Node-Edge-A, Peer-SID-ISP-1, Node-Core-1, Node-Core-2, ... , Node-8.8
PCEP sends SR Policy to Edge Router A:
  Binding SID: 40001
  Segment List: [16101, 32001, 16105, 16110, ...]
Edge Router A steers traffic:
  Match: 8.8.8.8/32 → Binding SID 40001
  Forward: [16101, 32001, 16105, ...]
```

☒ Complete Traffic Flow Example

Scenario: User at Edge-A wants to reach Google (8.8.8.8)

```
Step 1: Edge-A receives packet (8.8.8.8)
Step 2: ODN template matches → Headend requests path from PCE
Step 3: PCE computes optimal path (cost + latency):
  [Edge-A, Peer-ISP-1, Core-1, Core-5, Core-10, Edge-B, 8.8.8.8]
Step 4: PCEP programs SR Policy into Edge-A:
  Binding SID: 40001
  Segment List: [16101 (Edge-A), 32001 (Peer-ISP-1), 16105 (Core-1), ...]
Step 5: Edge-A steers traffic using Binding SID 40001
Step 6: Core forwards using top label (ECMP-aware Node SIDs)
Step 7: Edge-B receives, removes SR stack, forwards to 8.8.8.8
```

Result:

- ✓ Traffic via ISP-1 (cheaper, not ISP-2)
- ✓ BGP-free core (core only runs OSPF/IS-IS)
- ✓ <50ms failover (TI-LFA if Core-1 fails)
- ✓ Centralized TE (PCE computed optimal path)

☒ Benefits Achieved

Requirement	Solution
Egress optimization	BGP EPE (peer-level control)
Centralized TE	PCE (global optimization)
Network visibility	BGP-LS (topology to controller)
BGP-free core	SR-MPLS (IGP distributes SIDs)
Fast failover	TI-LFA (<50ms, no RSVP-TE)

Requirement	Solution
Automation	ODN (automatic SR Policy)

☒ CCDE Design Scenario 2: Complete Migration from MPLS/LDP to SR

☒ Migration Scenario Overview

Company: Enterprise Service Provider (ESP)

Current Network: MPLS with LDP + RSVP-TE

Network Size: 300 routers, 50 edge routers

Migration Goals:

- No service disruption during migration
- Gradual rollout (edge → core)
- Rollback capability if issues
- Full SR after migration (remove LDP/RSVP)

☒ Migration Strategy (5 Phases)

Phase 1: Pre-Migration Planning (Weeks 1-2)

Tasks:

1. Document current network:

└─ LDP sessions (300 routers × avg 4 peers = 1200 sessions)

└─ RSVP-TE tunnels (5000 tunnels)

└─ Label assignments (static labels, LDP labels)

└─ Traffic patterns (baseline utilization)

2. Define SR parameters:

└─ SRGB: [16000-23999] (8000 labels)

└─ Node SID mapping: Router ID → SID index

└─ Adj SID strategy (dynamic vs static)

3. Create migration playbook:

└─ Per-router config templates

└─ Rollback procedures

└─ Testing checklist

└─ Monitoring scripts

Node SID Mapping Table:

Router ID	Node SID Index	Actual Label
-----	-----	-----
10.0.0.1 (Core-1)	101	16101

```
10.0.0.2 (Core-2) | 102          | 16102
10.0.0.10 (Edge-1) | 201          | 16201
10.0.0.11 (Edge-2) | 202          | 16202
...
```

Phase 2: Enable IGP SR Extensions (Weeks 3-4)

Action: Enable SR on all IGP routers (core + edge) but keep LDP running

ISP-IS SR Configuration (All Routers):

```
router isis 1
  is-type level-2-only
  net 49.0001.0000.0000.0001.00
  !
  address-family ipv4 unicast
  !
  segment-routing
    sr-enabled
    !
    srgb
      base-index 16000
      range 8000
    !
    !
    node-sid
      prefix 10.0.0.1/32 ! Loopback
      index 101          ! Node SID index
      no-advertise       ! Don't advertise yet (Phase 3)
    !
  !
  !
  !
  !
```

What's Running After Phase 2:

Router: Core-1
✓ OSPF/IS-IS (IGP)
✓ SR extensions enabled
✓ SRGB configured [16000-23999]
✗ LDP still running (unchanged)
✗ RSVP-TE still running (unchanged)
✗ Node SIDs not advertised yet

Verification:

```
# Verify SR is enabled
show isis segment-routing summary
```

```
# Verify SRGB configured
show isis segment-routing srgb

# Verify LDP still running (should see sessions)
show ldp session
```

Phase 3: Enable SR on Edge Routers (Weeks 5-8)

Action: Enable SR on edge routers first (50 routers), advertise Node SIDs

Edge Router SR Configuration:

```
router isis 1
 address-family ipv4 unicast
   segment-routing
     sr-enabled
     !
   node-sid
     prefix 10.0.0.10/32 ! Edge-1 loopback
     index 201
     advertise           ! NOW advertise Node SID
     !
   !
 !
 !
 !

router ldp
 !
 ! Keep LDP running (parallel with SR)
 !
```

What's Running After Phase 3:

Edge Routers (50 routers):
✓ LDP running (existing)
✓ SR enabled (new)
✓ Node SIDs advertised (IGP)
✓ SR-LDP interworking enabled

Core Routers (250 routers):
✓ LDP running (existing)
✓ SR extensions enabled (Phase 2)
✗ Node SIDs NOT advertised yet
✗ LDP still primary (SR backup)

SR-LDP Interworking Configuration:

```
router ldp
!
segment-routing
interworking enable    ! LDP ↔ SR mapping
!
!

router segment-routing
!
ldp
interworking enable    ! SR ↔ LDP mapping
!
!
```

How SR-LDP Works:

Packet from Edge-1 (SR) to Core-1 (LDP-only in Phase 3):

Edge-1:

Push SR label [16101] (Node SID for Core-1)

Edge-1 (SR-LDP boundary):

Maps SR label 16101 → LDP label 1001

Pushes LDP label [1001]

Core-1 (LDP-only):

Processes LDP label 1001

Forwards to Core-1 destination

Phase 4: Enable SR on Core Routers (Weeks 9-16)

Action: Enable SR on core routers (250 routers), advertise Node SIDs gradually in batches

Batch Migration Plan (25 routers per batch):

```
Batch 1: Core-1 to Core-25 (Week 9-10)
Batch 2: Core-26 to Core-50 (Week 11-12)
Batch 3: Core-51 to Core-75 (Week 13-14)
...
Batch 10: Core-226 to Core-250 (Week 17-18)
```

Core Router SR Configuration (per batch):

```
router isis 1
address-family ipv4 unicast
segment-routing
sr-enabled
!
```



```
node-sid
prefix 10.0.0.1/32 ! Core-1 loopback
index 101
advertise          ! NOW advertise Node SID
!
!
!
!
```

Traffic Steering During Phase 4:

Option 1: LDP Primary, SR Backup (conservative)

- LDP still primary forwarding
- SR ready but not used yet
- Rollback: just disable SR (LDP still working)

Option 2: SR Primary, LDP Backup (aggressive)

- SR becomes primary forwarding
- LDP still running (rollback path)
- Rollback: disable SR, LDP takes over

Recommended: Option 1 (conservative) for first 3 batches, then Option 2

Verification After Each Batch:

```
# Verify Node SID advertised
show isis prefix-sid 10.0.0.1/32

# Verify SR label in forwarding
show mpls forwarding-table label 16101

# Verify LDP still working (rollback path)
show ldp session

# Verify traffic flow
ping mpls ipv4 10.0.0.1 ! LSP ping for SR
```

Phase 5: Remove LDP and RSVP-TE (Weeks 17-20)

Action: After all routers migrated to SR, remove LDP and RSVP-TE

How to Verify Migration Complete:

```
# Check all Node SIDs are advertised
show isis prefix-sid summary ! Should see 300 entries

# Check SR forwarding is working
show mpls forwarding-table ! Should see SR labels only

# Check LDP sessions (should be 0 or minimal)
show ldp session ! Should be empty or shutdown
```

```
# Check RSVP-TE tunnels (should be 0)
show mpls traffic-eng tunnels ! Should be empty
```

Remove LDP (All Routers):

```
no router ldp
```

Remove RSVP-TE (All Routers):

```
router isis 1
  no address-family ipv4 unicast
  mpls traffic-eng
!
!

no router mpls traffic-eng
```

Final State (After Phase 5):

Entire Network (300 routers):
✓ SR-MPLS primary (only protocol)
✓ Node SIDs advertised (all 300)
✓ TI-LFA enabled (<50ms failover)
✓ Flex-Algo 128 (low-latency paths)
✗ LDP removed (migrated)
✗ RSVP-TE removed (migrated)

☒ Complete Migration Rollback Plan

If Issue Detected in Any Phase:

Rollback Steps:

1. Disable SR advertisement:

```
router isis 1
  address-family ipv4 unicast
  segment-routing
  node-sid
  prefix 10.0.0.1/32
  no-advertise ! STOP advertising Node SID
!
!
!
!
!
```

2. Keep LDP running (already active):

```
router ldp
! (unchanged)
!
!
!
```

3. Verify LDP still forwarding:

```
show ldp session ! Should see sessions
show ldp interface ! Should see interfaces
```

4. Test traffic:

```
ping 10.0.0.1 ! Verify reachability
show mpls ldp forwarding ! Verify LDP labels working
```

Rollback Time: <15 minutes per router

☒ Migration Timeline Summary

Phase	Duration	Routers	What's Changed	Risk Level
1. Planning	2 weeks	0	Documentation, playbook	None
2. IGP SR	2 weeks	300	SR extensions enabled	Low
3. Edge SR	4 weeks	50	Node SIDs advertised, SR-LDP interworking	Medium
4. Core SR	8 weeks	250	Node SIDs advertised (batches)	Medium-High
5. Remove LDP/RSVP	4 weeks	300	LDP + RSVP removed	Low (if verified)
Total	20 weeks	300	Full SR	-

☒ Key Success Criteria

Criteria	How to Verify
No service disruption	Ping tests during each phase, monitoring shows 0% packet loss
Rollback capability	Test rollback procedure before each phase
All Node SIDs advertised	show isis prefix-sid summary = 300 entries
SR forwarding working	show mpls forwarding-table shows SR labels only
TI-LFA enabled	show segment-routing ti-lfa shows backup paths
LDP/RSVP removed	show ldp session=empty, show mpls traffic-eng tunnels=empty

❏ Common Migration Issues and Solutions

Issue	Root Cause	Solution
LDP sessions drop	SR-LDP interworking misconfig	Check <code>interworking enable</code> on both sides
Node SID not advertised	IGP not configured correctly	Verify <code>advertise</code> keyword on Node SID
Traffic loss during batch	SRGB mismatch between routers	Ensure all routers use same SRGB [16000-23999]
TI-LFA not working	Backup path not computed	Verify <code>segment-routing fast-reroute enabled</code>
SR labels not in FIB	CEF not updated	<code>clear mpls forwarding</code> or wait for CEF sync

❏ CCDE Design Scenario 3: Service Chaining with Segment Routing

❏ Scenario Overview

Company: Cloud Service Provider

Requirement: Force customer traffic through **security services** before reaching cloud:

1. **Firewall** (inspect traffic)
2. **IDS/IPS** (detect attacks)
3. **WAN Optimizer** (compress data)

Network: 100 routers, 3 service nodes (Firewall, IDS, WAN Optimizer)

❏ Service Chaining Design

Service Node Configuration

Service Nodes:

- └─ Firewall: 10.1.1.1/32, Node SID 16301
- └─ IDS/IPS: 10.1.1.2/32, Node SID 16302
- └─ WAN Opt: 10.1.1.3/32, Node SID 16303

Service Node Configuration (All 3 nodes):

```
router isis 1
 address-family ipv4 unicast
   segment-routing
     sr-enabled
   !
 node-sid
   prefix 10.1.1.1/32 ! Firewall loopback
   index 301
```

```
advertise
!
!
!
!
```

SR Policy for Service Chaining

Headend Router (Edge-1) Configuration:

```
router segment-routing
policy
  name SERVICE-CHAIN-1
  color 100 ! Service chain color
  endpoint 10.2.2.1 ! Final destination (cloud)
  !
  segment-list PRIMARY
    segment 16101 ! Edge-1 (self)
    segment 16301 ! Firewall
    segment 16302 ! IDS/IPS
    segment 16303 ! WAN Optimizer
    segment 16401 ! Cloud-Edge router
  !
!
!

router bgp 65000
neighbor 10.2.2.1
address-family ipv4 unicast
  next-hop-self ! Route to cloud
!
segment-routing
  policy SERVICE-CHAIN-1 ! Steer traffic to service chain
!
!
!
```

Traffic Flow

User → Edge-1 → [16301] → Firewall → [16302] → IDS → [16303] → WAN Opt → [16401] → C:

Step-by-step:

1. Edge-1 receives packet (to 10.2.2.1)
2. BGP matches next-hop → SR Policy SERVICE-CHAIN-1
3. Edge-1 pushes segment stack: [16301, 16302, 16303, 16401]
4. Firewall (16301) receives, pops 16301, processes traffic
5. Firewall forwards with [16302, 16303, 16401]
6. IDS (16302) receives, pops 16302, processes traffic
7. IDS forwards with [16303, 16401]
8. WAN Opt (16303) receives, pops 16303, compresses

9. WAN Opt forwards with [16401]
10. Cloud-Edge (16401) receives, removes stack, forwards to cloud

Result:

- ✓ Traffic goes through all 3 services in order
- ✓ No additional protocol needed (just SR segment list)
- ✓ Centralized control (controller can modify service chain)

☒ Summary: All CCDE Design Scenarios Covered

#	Scenario	Key Concepts
1	Provider Network with SR-EPE, BGP-LS, PCE	BGP EPE, BGP-LS, PCE, ODN, SDN
2	Complete Migration from MPLS/LDP to SR	SR-LDP Interworking, 5-phase migration, rollback
3	Service Chaining with SR	Service Node SID, Binding SID, SR Policy
4	Flex-Algo for Low-Latency Paths	Flex-Algo 128, custom IGP metric
5	Seamless MPLS Multi-Domain	BGP Prefix SID, ABR, multi-IGP

Would you like **configuration deep-dive** for any specific scenario (e.g., *"Full BGP-LS + PCE setup on Cisco IOS XR"*), **troubleshooting guide** for migration issues, or **additional scenarios** (e.g., *"5G Mobile Network with SRv6"*, *"Data Center Interconnect with SR-BGP"*).